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AI/ML Engineer

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Indian Institute of Information Technology, Allahabad

Objective:

To work with an organization, which can provide me constant learning, leading to intellectual growth, enhance my creative skills and to achieve top cadre by setting benchmarks both at organizational and personal front and making positive contributions towards the organization.

Highlights:

- About **7+ years** of experience as **Artificial Intelligence Machine Learning** which involves Application Development in **Artificial Intelligence Machine Learning, Python, Data Science, Django, FastAPI and Google Cloud Platform.**
- **Currently** working as, a **AIML Engineer** with **Springer Nature Technology and Publishing Solutions, Pune,** developing enterprise-scale **Generative AI (GenAI)** solutions including Alternative Text Generation, Taxonomy Discovery, Ethics Compliance Validation, and Budget & Campaign Optimization platforms for global marketing teams across multiple regions. Specialized in designing and deploying scalable Python-based AI applications leveraging **Large Language Models (LLMs)** such as OpenAI, Gemini, and other open-source and commercial foundation models. Experienced in building **Retrieval-Augmented Generation (RAG)** pipelines using **LangChain, vector databases, embeddings, prompt engineering, semantic search, and document retrieval frameworks**. Implemented AI observability and monitoring using **Langfuse, Grafana, and Prometheus** for performance tracking, prompt evaluation, cost optimization, and model analytics, integrating semantic search, embeddings, vector databases, and advanced prompt engineering techniques. Developed and deployed Google Workspace Add-ons using Google Apps Script and successfully published solutions on Google Marketplace.
- **Worked** as, a **Python Developer** with client side of **MPMKVVCL (Madhya Pradesh Madhya kshetra Vidyut Vitran Co. Ltd, Bhopal)** i.e**. State Power Generation Company** outsourced by **MDP INFRA India Pvt ltd, Bhopal** for the Commercial website development called as QCPortal for their **Vendor/ NABL/ TKC/ Solar facilitation, registrations, and LAB equipment (Transformers, Meters, Wire etc.)** onboarding since October 2021. Designed **machine learning solutions** including Theft Detection Models for theft analytics using NBS billing data. Built an end-to-end intelligent electric meter reading pipeline using **YOLO v7, Python and AI/ML Generative Vision-Language Models** (TrOCR), enabling automated optical character recognition and data extraction directly from meter display images. Developed REST APIs, data processing workflows, and automation pipelines to streamline operational processes.
- Worked as **Software Developer** with **CyberInfrastucture Pvt. Ltd, Indore** from March 2021 to October 2021 for building **Machine Learning application** for US client and Python code Israel Military Force. Developed an **AI/ML-powered** Risk Maturity Assessment Platform using Python, Streamlit, Django, and FastAPI. Built interactive web applications, secure user authentication workflows, and scalable **ML-powered APIs** for automated risk assessment, analytics, and report generation, enabling data-driven enterprise decision-making.
- Worked as **Python Programmer** with **Aspirify Enterprise Pvt. Ltd, New Delhi** from August 2020 to March 2021. Developed Django-based enterprise applications using Python, Celery, and Redis for scalable workflow automation and distributed processing. Built **AI/ML-powered** network scanning and vulnerability assessment solutions that automated risk detection, security analytics, and insight generation through

intelligent data-processing pipelines, enabling data-driven decision-making and operational efficiency.

- **Pursued** internship **as a Data Science Engineer** with **Innoviti Payment Solution PVT LTD*, *Bangalore** from February 2020 to May 2020. where I worked on data analysis, machine learning model development, feature engineering, and business intelligence solutions for payment and transaction datasets.
- Worked as **Python Developer** with **SNSE, New Delhi** from July 2017 to August 2019. developing data analytics solutions, web scraping systems, Python automation scripts, and microservice-based APIs. **Designed ETL workflows for large-scale data extraction, transformation, and processing, enabling business intelligence and reporting solutions.** Collaborated on multiple client projects involving backend development, data engineering, and process automation using Python technologies.
- Worked as **Software Trainee** at **CDAC, Jaipur** from August 2014 to July 2015 for building Java application named "Puzzle Game", Advanced Java application named "Crime File Management System" and Android application named "iDoc".
- During my **M. Tech from IIIT Allahabad**, did projects on DNA & Genome Sequencing, Pattern Matching, Olfaction Detection using Machine Learning and Data Science.

Academic Qualifications:

- ⇨ **Bachelor of Technology in Computer Science & Information Technology** Engineering from **Poornima College, Jaipur i**n 2016.
- ⇨ **Master of Technology in Information Technology** from **Indian Institute of Information Technology Allahabad, Uttar Pradesh** in 2019.
- ⇨ **Professional Certificate Program in Business Management (PCPBM)** from **Indian Institute of Management, Kozhikode, Kerala** in 2022.

Technical Skills'

- ➢ **Operating Systems** : Microsoft Windows / Ubuntu(Linux) / MacOS
- ➢ **Languages** : PYTHON / HTML / C/C++ / Java / Android / Google Apps Script
- ➢ **Generative AI & LLMs** : Generative AI (GenAI), Large Language Models (LLMs), OpenAI, Gemini, Prompt Engineering
- ➢ **RAG & Knowledge Systems** : Retrieval-Augmented Generation (RAG), LangChain, Semantic Search, Embeddings, Vector Databases (Chroma DB, Pinecone)
- ➢ **Machine Learning & AI** : Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, OCR (TrOCR), Model Evaluation, AI Observability
- ➢ **Web Development Methods** : Django, Django REST Framework, Google Cloud Platform
- ➢ **Front end Technologies** : HTML5 / CSS / JavaScript
- ➢ **Version Control Technologies** : GIT / JIRA
- ➢ **Agile Development Tools** : JIRA / CONFLUENCE
- ➢ **Databases** : PostgreSQL, MySQL, Mongo DB, SQL Server, Oracle
- ➢ **Other tools** : Celery, Redis, Docker, Jenkins
- ➢ **Testing Framework** : Pytest / Sonar Cube
- ➢ **Software Packages** : MS Word, MS Excel, MS PowerPoint
- ➢ **Protocols** : TCP/IP, HTTP/HTTPS, SMTP
- ➢ **Methodologies** : Agile, Scrum and Waterfall
- ➢ **Monitoring** : Langfuse, MLflow, Grafana, Prometheus, Model Monitoring, Performance Evaluation, Cost Optimization

Certifications:

- ➢ Generative AI for Everyone course completion certificate from Coursera in 2024
- ➢ Machine Learning Specialization course completion certificate from Coursera in 2023

- Certificate of completion for Chatbot Builder Training from Yellow.ai Academy on 20 June 2022.
- Build a Data Science Web App with Streamlit and Python certificate from Coursera on 28 July 2020
- Introduction to Python certificate from Coursera on 20 June 2020
- Python Data Structures certificate from Coursera on 5 May 2020
- Python for Data Science Certificate from Intellipaat on 17 Jan 2020.
- Machine Learning certificate of training from Internshala Trainings with 83 percent marks from 15th Nov to 27th Dec 2019.
- Data Science course completion certificate from LearnVern on 21 June, 2019.
- Advance Java Certification from C-DAC, Jaipur in 2015. Java Certification from C-DAC, Jaipur in 2014.
- Diploma in Hardware and Maintenance from SIIT Computer Education in 2005.
- Diploma in Computer Aided Designing from Heights Computer Education in 2005.

Projects Details:

Organization – Springer Nature Technology and Publishing Solution (Pune)

Python and Machine Learning Application Development : (April 2023 – PRESENT):

Technology : **Python / Django / Machine Learning / NLP / Google Cloud Platform / Fast API / Apps Script / Python Script / Streamlit / JIRA / AGILE**

Duration : 38 months (till now)

Team Size : 12 members

Project 1 Descriptions: The aim is to provide the marketeers with a tool to test different marketing strategies and budget allocation. With the tool a marketeer is able to enter online touchpoints driven by marketing for specific parts of the user journey and in return get an understanding of the potential revenue impact.

Responsibilities : Create UI/UX for the tools using Django python and optimize and fine-tunned the model and plot region wise continental graphs for potential submitters needs. Budget & Campaign Optimization platforms for global marketing teams across multiple regions.

Project 2 Descriptions: Augmented Insights is an internal toll for editors that allows a human-machine handshake for augmented content creation with large language models.

Responsibilities : Create Google Docs Addon using Apps Script for Augmented Insights and deployed Google Workspace Add-ons using Google Apps Script and successfully published solutions on Google Marketplace.

Project 3 Descriptions: An external DB of EBMs from various journals to enable targeted and tracked communication.

Responsibilities : Data Extraction, Cleaning, Profiling of Jflow free text from XML from the server into CSV files, Merge new and old data, upload the data to SAP server.

Project 4 Descriptions: Explore alternative of LLM Vision model GPT4-Vision for identification of images which will be extracted from PDF and do the cost analysis for all the models.

Responsibilities : Build script to extract PDF from ISBN, extract images from PDF, find alternative of GPT4-Vision and deploy it on the server using FAST API for backend and streamlit for user end.

Project 5 Descriptions: Alternative Text generator for image from Journal and Books.

Responsibilities : Designed and developed an Gen AI-based system to automatically generate alternative (alt) text descriptions for scientific and academic images. Integrated vision-language models to extract meaningful insights from complex figures, charts, and diagrams and deploy it on the server using FAST API for backend and streamlit for user end. Implemented AI observability and monitoring using Langfuse, Grafana, and Prometheus for performance tracking, prompt evaluation and cost optimization.

Project 6 Descriptions: Taxonomy Finder (classify and map content into predefined taxonomies)

Responsibilities : Developed an intelligent system to automatically classify and map content into predefined taxonomies using Gen AI and machine learning techniques. The solution analyzes structured and unstructured data to accurately assign categories, subcategories, and hierarchical tags.

Project 7 Descriptions: Evaluate research documents for ethical considerations related to human and animal studies.
Responsibilities : Developed an Gen AI-based compliance system to evaluate research documents for ethical considerations related to human and animal studies. The system analyzes manuscripts, protocols, and reports to verify the presence of required ethical approvals, consent statements, and regulatory compliance.

Organization – Madhya Pradesh Madhya Kshetra Vidyut Vitran Co. Ltd outsourced by MDP INFRA PVT LTD. (Bhopal)

Transformer Onboarding for XMER Repairing and Quality Monitoring Portal for Madhya Pradesh Central Discomm:(: (October 2021 – March 2023)

Technology : **Python/ Machine Learning / Django / Django Rest Framework / GIT / HTML / JS / CSS / Bootstrap**
Duration : 18 months
Team Size : 6 members

Project 1 Descriptions: Transformer onboarding or repairment and tracking the details of vendors, contractors, and testing lab users to properly accommodate the sending and receiving of the transformer as per their xmer code. Document tracing and auto generated mail setup tasks. This app is handling 50000 users.

Responsibilities : Backend product development in Django.

Project 2 Descriptions: "**Quality Monitoring Portal**" is the commercial website for Registration of Vendors, NABL Accredited LABS, Officers of MPCZ distribution company, Turnkey Contractors. Central Purchase system module attached to QCPortal is for the payment management of Vendors.

Responsibilities: Backend Development with Django and PostgreSQL.

Checkout the project at https://qcportal.mpcz.in

Project 3 Descriptions: Electricity Theft Detection for MPCZ Company

Responsibilities: Developed a machine learning-based system to detect electricity theft and abnormal consumption patterns for MPCZ (Madhya Pradesh Central Zone) Company. Designed machine learning solutions including Theft Detection Models for theft analytics using NBS billing data. The solution analyzed historical consumption data to identify suspicious behavior, reduce revenue losses, and improve operational efficiency.

Project 4 Descriptions: intelligent electric meter reading for MPCZ Company

Responsibilities: Built an end-to-end intelligent electric meter reading pipeline using YOLO v7, Python and AI/ML Generative Vision-Language Models (TrOCR), enabling automated optical character recognition and data extraction directly from meter display images. Developed REST APIs, data processing workflows, and automation pipelines to streamline operational processes.

Organization – Cyberinfrastructure Private Ltd (Indore)

Python and Machine Learning Development: (March 2021 to October 2021)

Technology : **Python / Machine Learning / Streamlit / Fast API / JIRA**
Duration : 8 months
Team Size : 2 members

Project 1 Descriptions: Develop code modules using python for Israel Military Force outsourced to CIS, Indore.

Responsibilities : Created modular code using traditional Machine Learning algorithms application for

US client and Python code Israel Military Force

Project 2 Descriptions: Need to develop Risk Maturity Assessment System based on questionaries. We have to build a website in which corporate user can login, fill the details provided and do the risk maturity assessment for their organization and generate a report based on the questions asked.

Responsibilities : Developed an AI/ML-powered Risk Maturity Assessment Platform using Python, Streamlit, Django, and FastAPI. Built interactive web applications, secure user authentication workflows, and scalable ML-powered APIs for automated risk assessment, analytics, and report generation, enabling data-driven enterprise decision-making.

Organization – Aspirify Enterprise Pvt Ltd (New Delhi)

Network scanning: (Aug 2020 – March 2021)

Technology : **Python/Machine Learning/ Numpy/ Pandas/ MATPLOTLIB** and **MATLAB**
Duration : 8 months
Team Size : 6 members
ProjectType : **Defense and R&D**

Project Descriptions: Build a Django Web Project with Celery and Redis for parallel and fast network scanning to find vulnerabilities and gather the information across the network

Responsibilities : Build a Django python web application for network scanning using Python, Celery, and Redis for scalable workflow automation and distributed processing. Built AI/ML-powered network scanning and vulnerability assessment solutions that automated risk detection, security analytics, and insight generation through intelligent data-processing pipelines, enabling data-driven decision-making and operational efficiency.

Organization – Innoviti Payment Solution Pvt Ltd (Bangalore)

Network scanning: (February 2020 – May 2020)

Technology : **Python / NumPy / Pandas / MATPLOTLIB**
Duration : 4 months
Team Size : 1(Myself)
ProjectType : **Working as Intern as Data Science and Machine Learning Engineer**

Project Descriptions: Analysis the data of various bank's transaction which are intact with Innoviti payment solution

Responsibilities : Liaising with the Development team and help them in the development activities, working on PostgreSQL to interact with company's database, Writing the Queries on the PostgreSQL and Jupyter notebook to fetch the data from DB and worked on data analysis, machine learning model development, feature engineering, and business intelligence solutions for payment and transaction datasets.

Organization – SNSE (New Delhi)

Python Development: (July 2017 – August 2019)
Built AI-powered automation solutions leveraging Python, Artificial Intelligence Machine Learning, NLP, Computer Vision, and data engineering techniques, laying the foundation for modern Generative AI and LLM-based application development.

Technology	: **Python, Machine Learning, Data Analytics, NumPy, Pandas, Matplotlib, OpenCV, NLP, Django, REST APIs, MySQL, AWS, Web Scraping, HTML, Bootstrap, ROS, Git**
Duration	: 26 months
Team Size	: 3 members

Project 1 Descriptions: Knowledge Management & Question-Answer Portal. Developed a knowledge aggregation and intelligent Q&A platform that enabled users to browse domain-specific content, contribute question-answer pairs, and manage knowledge repositories. Implemented backend APIs and data management features to support content discovery and retrieval.

Responsibilities : Developed Python-based backend services and REST APIs. Built responsive UI/UX components using HTML and Bootstrap. Designed database schemas and optimized data retrieval workflows. Implemented content categorization and search functionalities. Assisted in NLP-based text processing for content organization and knowledge extraction.

Project 2 Descriptions: AI-Based Victim Detection System for Rescue Robots. This project presents an analysis of the face detection of victims. The focus is mainly on image processing for autonomous robots operating for detecting victims in crisis areas. The purpose of this robot to be applied in a rescue scenario to detect the victim. Its focuses on one of the main objectives of rescue missions is to rescue the disaster victims. Of course, in some situations, it is too risky to send human agents in an unstable environment.

Responsibilities : Developed computer vision pipelines using Python and OpenCV. Implemented face detection and image analysis algorithms. Processed and enhanced image datasets for model training and evaluation. Integrated vision modules with robotic systems using ROS. Performed model evaluation and accuracy optimization.

Project 3 Descriptions: Recommendation Engine using Machine Learning. Developed collaborative filtering and content-based recommendation systems to generate personalized product recommendations. Applied machine learning and data analytics techniques to identify user preferences and product similarities.

Responsibilities : Built recommendation algorithms using Python and Machine Learning. Performed data cleaning, preprocessing, and feature engineering using Pandas and NumPy. Conducted exploratory data analysis (EDA) and data visualization. Evaluated recommendation accuracy and performance metrics. Generated business insights through predictive analytics.

Project 4 Descriptions: Employee & Leave Management System. Developed a Django-based enterprise web application for employee management and leave administration. The platform streamlined HR operations through automated workflows and centralized employee information management.

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Responsibilities : Designed and developed Django-based web applications. Created REST APIs and backend business logic. Managed employee records, leave workflows, and reporting modules. Implemented database design and optimization using MySQL. Integrated authentication and role-based access control features.

Project 5 Descriptions: Python Automation, Data Analytics & API Development. Delivered multiple client projects involving data analytics, web scraping, API development, process automation, and business intelligence solutions. Developed scalable Python applications for data ingestion, transformation, and reporting.

Responsibilities : Developed Python automation scripts for data extraction and processing. Built RESTful Microservices and backend APIs. Created web scraping solutions for large-scale data collection. Performed data analysis, visualization, and reporting. Developed ETL pipelines for structured and unstructured data. Worked closely with clients to deliver customized AI/ML and data-driven solutions.

ACHIEVEMENTS:

- **Certificate of paper to Manuscript entitled "PREDICTING SMELL PERCEPTION FROM MOLECULAR DESCRIPTORS USING MACHINE LEARNING APPROACH" in Institute of Electrical and Electronics Engineers held at 2020 International Conference Engineering and Telecommunication (En&T 2020), Dolgoprudny, Russia 25-26 November 2020 published in IEEE Catalog**

Number- CFP20X41-POD, ISBN- 978-1-7281-8830-0, page 331.

- **Certificate of paper to Manuscript entitled “Mobile Virtual Class” in Imperial Journal of Interdisciplinary Research (IJIR) in 2017 vol-3, issue-4.**
- **Certificate of best paper to Manuscript entitled “iDoc” in National Conference on Smart Computation and Technology (NCSCT) held at Poornima Institute of Engineering & Technology, Jaipur on April 8th-9th, 2016 published in International Journal of Computer Trends and Technology (IJCTT)V35(5).**
- **Certificate of paper Manuscript entitled “Brain Chip as the solution for the development of society” at National Conference on Recent Innovations in Engineering (NCRIE) held at Apex Group of Colleges, Jaipur & Kota during 26th-27th April, 2014 published in International Journal of Engineering, Management & Sciences (18).**
- **Certification for appreciation for being an organizer in “International Conference on Bioinformatics and System Biology - 2018 at IIITA on October 26th-28th,2018.**
- Selected two times for **State Level for Badminton** championship
- Selected for **campus selection of Foot Ball Organized** by Govt of Madhya Pradesh.
- GATE **2016/2017** qualified.

Declaration:

I hereby declare that the above written particulars are true to the best of my knowledge and belief.

Place: Pune, Maharashtra, India

Date: 12-06-2026
(Anupam Vishwakarma)